

Internship Report

On

**“MODELLING OF BLOOD FLOW THROUGH
DEFORMABLE MICROVESSELS”**

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DECLARATION

I, Nirnoy Barma, hereby declare that the research work titled “Modelling of Blood Flow Through Deformable MicroVessels” is my original work and has been conducted jointly and independently as part of our academic internship requirements.

All sources of information, mathematical formulations, references, and collaborative research frameworks utilized in this report have been appropriately acknowledged. Full credit has been given to the original authors and literature as per university academic and ethical standards. Any data, experimental reference curves, or analytical concepts derived from the work of others are explicitly cited to ensure proper attribution.

I affirm that this research has not been previously submitted for evaluation or grading in any other course, department, or institution, and has been compiled solely for academic completion purposes.

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1 Introduction

The rapid advancement of bio microfluidic technologies and lab-on-a-chip devices has revolutionized medical diagnostics, targeted drug delivery, and physiological modeling [1]. Central to these microsystems is the hydrodynamic prediction of complex biological suspensions like blood, which exhibit highly non-Newtonian characteristics under confinement. In the microvasculature, the vast majority of flow resistance occurs in small diameter vessels ($< 300 \mu m$) [2]. At this scale, blood flow is governed by two classic, interconnected phenomena: the Fåhræus effect (the reduction of tube hematocrit relative to discharge hematocrit) and the Fåhræus-Lindqvist effect (the decrease in apparent blood viscosity with decreasing vessel radius) [3, 4].

The physical basis for these phenomena is Haynes' marginal zone theory [5], which describes the shear-induced axial migration of red blood cells (RBCs) toward the high-velocity centerline. This creates a two-phase, core-annulus flow architecture consisting of a highly concentrated, erythrocyte-rich central core and a peripheral, cell-free plasma layer (CFL) adjacent to the wall. Traditional mathematical continuum models capture this multi-layer behavior by applying local momentum balances across both zones [6, 7]. However, a major limitation of classical frameworks including those by Chebbi [6], Ascolese et al. [7], and Farina et al. [8] is the rigid-tube assumption, which treats the confining channel as a perfectly unyielding cylinder.

In physiological reality and advanced polymeric microchannels (such as PDMS), the boundaries are inherently compliant [1, 9]. Low-Reynolds-number fluid-structure interactions (FSIs), known as soft hydraulics, reveal that hydrodynamic pressure induces local elastic deformations of the wall, altering the cross-sectional area, modifying the axial pressure gradient, and generating non-linear flow rate-pressure drop relationships [9, 10]. While engineering models have coupled lubrication theory with structural mechanics for single-phase non-Newtonian flows [10, 11], a coupled framework resolving multi-phase microvascular cell migration alongside a deformable boundary layer remains an open frontier.

This research bridges this gap by presenting a comprehensive rheological model that couples the Fåhræus-Lindqvist effect with a deformable boundary layer. Moving beyond single-phase approximations, we model a two-phase core-annulus flow where the concentrated RBC core is treated as a non-Newtonian fluid governed by the Carreau constitutive model to capture its shear-thinning behavior, while the peripheral plasma annular region is treated as a Newtonian fluid governed by the classic Navier-Stokes equations. Crucially, the channel is lined with a compliant, deformable endothelial glycocalyx layer (EGL) - a soft, porous macromolecular brush coat. Under the lubrication approximation for a slender channel, the hydrodynamic pressure varies solely along the axial coordinate, yielding $p = p(z)$, which directly drives the local elastic compression of the EGL.

To approach this problem, we systematically couple the multi-phase fluid dynamics with the structural mechanics of the deformable EGL. First, the axial velocity profiles are analytically derived by integrating the zone-specific momentum equations, ensuring continuity of velocity and shear stress at the moving core-annulus interface, paired with an elastic displacement or hydrodynamic boundary condition at the compressed EGL surface. Second, integrating these velocity distributions yields a generalized volumetric flow rate relation as a function of the local deformed channel radius $R(z)$. Third, this relation is coupled to a structural mechanism where local EGL deformation $u_r(z) = R(z) - R_0$ is proportional to the local gauge pressure $p(z)$. This yields a non-linear, first-

order ordinary differential equation (ODE) for the axial pressure distribution $p(z)$. We resolve this governing system using a regular perturbation expansion in the dimensionless Carreau number Cu for weak non-linearities, alongside a robust numerical fourth-order Runge-Kutta scheme paired with a shooting method. This integrated approach allows us to precisely quantify how boundary compliance and non-Newtonian rheology modulate apparent viscosity, centerline velocity, core hematocrit levels, and mechanical energy dissipation.

2 Theory

2.1 Introduction and Physical Model

To accurately capture the microrheological behavior of blood flow in small-diameter vessels specifically modeling the Fåhræus and Fåhræus-Lindqvist effects we employ a two-phase localized continuum framework based on Haynes' marginal zone theory. Blood flow within a capillary of radius R is partitioned into two distinct axisymmetric concentric regions:

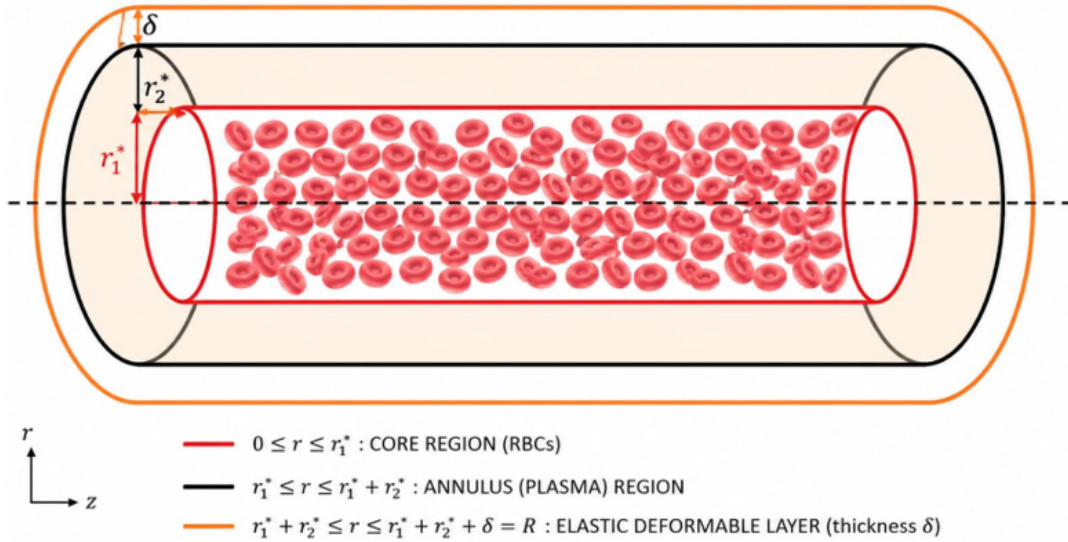


Figure 1: Two-phase core-annulus flow architecture with a compliant boundary layer.

- **A Core Region** ($0 \leq r \leq r_1^*$): Comprising a concentrated, homogeneous suspension of red blood cells (RBCs). Given the macromolecular aggregation and alignment of erythrocytes under shear, the core fluid exhibits pronounced shear-thinning behavior, which we mathematically describe using the Carreau constitutive model.
- **An Annular Peripheral Region** ($r_1^* \leq r \leq r_1^* + r_2^*$): Representing a cell-depleted plasma layer. This cell-free marginal zone is treated as an isotropic Newtonian fluid layer.

The total vessel radius is bounded by the structural limits of the domains such that $R = r_1^* + r_2^* + \delta$ where δ , signifies the thickness of an outer bounding layer (e.g., an elastic deformable endothelial surface glycocalyx layer).

2.2 Mathematical Formulation & Governing Equations

The dynamic local viscosity of the non-Newtonian core phase, $\mu_c(\dot{\gamma})$, is governed by the four-parameter Carreau model to evaluate structural transitions over the complete spectrum of shear rates:

$$\mu_c(\dot{\gamma}) = \mu_\infty + (\mu_0 - \mu_\infty)[1 + (\lambda\dot{\gamma})^2]^{\frac{n-1}{2}} \quad (2.1)$$

Where μ_0 is the zero-shear-rate viscosity plateau, μ_∞ is the infinite-shear-rate viscosity infinity limit, λ is a characteristic relaxation time constant of the fluid, and n is the dimensionless behavior index ($n < 1$ for shear-thinning media).

Conversely, the peripheral plasma phase in the annulus maintains a constant Newtonian viscosity:

$$\mu_a = \text{constant} \quad (2.2)$$

2.3 Hydrodynamic Systems Configurations

Assuming steady-state, incompressible, fully developed laminar flow along the axial direction (z), the equations of motion reduce to a one-dimensional form of the Cauchy momentum equation. The fundamental force balance in cylindrical coordinates across any arbitrary fluid layer is given by:

$$\frac{\partial p}{\partial z} = \frac{1}{r} \frac{d}{dr}(r\tau_{rz}) \quad (2.3)$$

Integrating once with respect to the radial coordinate r under the required physical condition of finite momentum at the centerline axis ($\tau_{rz} = 0$ at $r = 0$) yields the explicit radial shear stress distribution:

$$\tau_{rz}(r) = \frac{r}{2} \frac{dp}{dz} \quad (2.4)$$

3 Core and Annular Core Solutions ($0 \leq r \leq r_1^* + r_2^*$)

3.1 Core Region Non-Dimensionalization ($0 \leq r \leq r_1^*$)

To generalize the fluid configurations within the core domain, we construct the following non-dimensional system variables:

$$\bar{r} = \frac{r}{R}, \quad \bar{v}_c = \frac{v_c}{U_c}, \quad \bar{\mu}_c = \frac{\mu_c}{\mu_0}, \quad \bar{p} = \frac{p}{P_c}, \quad \bar{z} = \frac{z}{L} \quad (3.1)$$

Substituting these scaling mappings into the core momentum balance results in the non-dimensional equation of motion:

$$\bar{\mu}_c \frac{d\bar{v}_c}{d\bar{r}} = \frac{\bar{r}}{2} \frac{d\bar{p}}{d\bar{z}} \quad (3.2)$$

The scalar kinematics yield an expression for the localized magnitude of the shear rate $\dot{\gamma}$:

$$\dot{\gamma} = \left| \frac{\partial v_c}{\partial r} \right| = \frac{U_c}{R} \left| \frac{d\bar{v}_c}{d\bar{r}} \right| \quad (3.3)$$

We group the characteristic time metrics by defining the dimensionless Carreau number (Cu), which represents the ratio of the fluid relaxation time to the characteristic timescale of the bulk flow:

$$Cu = \frac{\lambda U_c}{R} \quad (3.4)$$

This identification directly leads to the non-dimensionalized shear rate group:

$$\lambda \dot{\gamma} = Cu \left| \frac{d\bar{v}_c}{d\bar{r}} \right| \quad (3.5)$$

By introducing the asymptotic viscosity scale parameter $\beta = \frac{\mu_\infty}{\mu_0}$, the non-dimensional Carreau profile updates to:

$$\bar{\mu}_c = \beta + (1 - \beta) \left[1 + Cu^2 \left(\frac{d\bar{v}_c}{d\bar{r}} \right)^2 \right]^{\frac{n-1}{2}} \quad (3.6)$$

Isolating the localized velocity gradient field explicitly produces:

$$\frac{d\bar{v}_c}{d\bar{r}} = \frac{\bar{r}}{2} \frac{d\bar{p}}{d\bar{z}} \left[\beta + (1 - \beta) \left[1 + Cu^2 \left(\frac{d\bar{v}_c}{d\bar{r}} \right)^2 \right]^{\frac{n-1}{2}} \right]^{-1} \quad (3.7)$$

3.2 Perturbation Analytical Scheme (Weak Carreau Variations)

For microvascular configurations experiencing highly localized shear or low characteristic flow states ($Cu \ll 1$), the systemic non-linearities can be expanded using a binomial series representation. Letting $G = \frac{d\bar{v}_c}{d\bar{r}}$, the dimensionless viscosity profile expands to:

$$\bar{\mu}_c = 1 + \frac{(1 - \beta)(n - 1)}{2} Cu^2 G^2 + \frac{(1 - \beta)(n - 1)(n - 3)}{8} Cu^4 G^4 + \mathcal{O}(Cu^6) \quad (3.8)$$

To invert this operator into the tracking momentum frameworks, a consecutive binomial expansion yields:

$$(\bar{\mu}_c)^{-1} = 1 - \frac{(1 - \beta)(n - 1)}{2} G^2 Cu^2 + \left[\frac{(1 - \beta)^2(n - 1)^2}{4} - \frac{(1 - \beta)(n - 1)(n - 3)}{8} \right] G^4 Cu^4 \quad (3.9)$$

We establish a regular perturbation series expansion of the internal velocity gradient where $G = G_0 + Cu^2 G_1 + \dots$. Resolving to leading-order ($\mathcal{O}(1)$) yields the base structural solution:

$$G_0 = \frac{\bar{r}}{2} \frac{d\bar{p}}{d\bar{z}} \quad (3.10)$$

Substituting Eq. (3.10) back into Eq. (3.8) yields the regularized viscosity field expression:

$$\begin{aligned} \bar{\mu}_c = 1 + \frac{(1 - \beta)(n - 1)}{2} Cu^2 \frac{\bar{r}^2}{4} \left(\frac{d\bar{p}}{d\bar{z}} \right)^2 \\ + \frac{(1 - \beta)(n - 1)(n - 3)}{8} Cu^4 \left(\frac{d\bar{p}}{d\bar{z}} \right)^4 \frac{\bar{r}^4}{16} + \mathcal{O}(Cu^6) \end{aligned} \quad (3.11)$$

We collect grouping definitions for the spatial geometric distributions by defining the auxiliary variables H_1 and H_2 :

$$H_1 = \frac{(1-\beta)(n-1)}{8} \left(\frac{d\bar{p}}{d\bar{z}} \right)^2 \bar{r}^2 = C\bar{r}^2, \quad \text{where} \quad C = \frac{(1-\beta)(n-1)}{8} \left(\frac{d\bar{p}}{d\bar{z}} \right)^2 \quad (3.12)$$

$$H_2 = \frac{(1-\beta)(n-1)(n-3)}{128} \left(\frac{d\bar{p}}{d\bar{z}} \right)^4 \bar{r}^4 = D\bar{r}^4, \quad \text{where} \quad D = \frac{(1-\beta)(n-1)(n-3)}{128} \left(\frac{d\bar{p}}{d\bar{z}} \right)^4 \quad (3.13)$$

The relative operational fluid properties simplify into clean expressions:

$$\bar{\mu}_c = 1 + H_1 Cu^2 + H_2 Cu^4 + \mathcal{O}(Cu^6) \quad (3.14)$$

$$\frac{1}{\bar{\mu}_c} = 1 - H_1 Cu^2 + (H_1^2 - H_2) Cu^4 \quad (3.15)$$

Substituting Eq. (3.15) back into the definition of G , we perform a direct spatial integration with respect to $\bar{r}$:

$$\bar{v}_c = \frac{1}{2} \frac{d\bar{p}}{d\bar{z}} \left(\frac{\bar{r}^2}{2} - \frac{C\bar{r}^4}{4} Cu^2 + \frac{(C^2 - D)\bar{r}^6}{6} Cu^4 \right) + C_1 \quad (3.16)$$

Expanding the integration terms explicitly yields the analytical solution for velocity within the core region:

$$\begin{aligned} \bar{v}_c = & \frac{1}{4} \frac{d\bar{p}}{d\bar{z}} \bar{r}^2 - \frac{1}{64} (1-\beta)(n-1) \left(\frac{d\bar{p}}{d\bar{z}} \right)^3 \bar{r}^4 Cu^2 \\ & + \frac{(1-\beta)(n-1)[6(1-\beta)(n-1) - (n-3)]}{1536} \left(\frac{d\bar{p}}{d\bar{z}} \right)^5 \bar{r}^6 Cu^4 + C_1 \end{aligned} \quad (3.17)$$

3.3 Annular Layer Region Solution ($r_1^* \leq r \leq r_1^* + r_2^*$)

Within the marginal cell-free layer, the plasma behaves as a classical Newtonian fluid. The axial momentum balance equation simplifies to:

$$\frac{\partial p}{\partial z} = \frac{\mu_a}{r} \frac{d}{dr} \left(r \frac{dv_a}{dr} \right) \quad (3.18)$$

Integrating Eq. (3.18) once with respect to the radius r provides the shear gradient layout for the Newtonian phase:

$$\mu_a \frac{dv_a}{dr} = \frac{r}{2} \frac{dp}{dz} \quad (3.19)$$

To systematically link the behavior of the annular layer with the core domain, we enforce identical scaling definitions along with a characteristic visco-capillary pressure scale (P_c) and viscosity ratio (μ_r):

$$\bar{r} = \frac{r}{R}, \quad \bar{v}_a = \frac{v_a}{U_c}, \quad \bar{p} = \frac{p}{P_c}, \quad \bar{z} = \frac{z}{L}, \quad \mu_r = \frac{\mu_a}{\mu_0}, \quad P_c \sim \frac{\mu_0 U_c L}{R^2} \quad (3.20)$$

Performing a spatial integration under this non-dimensional optimization framework yields the velocity distribution profile for the outer annular layer:

$$\bar{v}_a = \frac{1}{4\mu_r} \frac{d\bar{p}}{d\bar{z}} \bar{r}^2 + C_2 \quad (3.21)$$

4 Boundary Condition Closure and Regional Velocity Profiles

4.1 Plasma Layer Boundary Closure ($\bar{r} = \frac{r_1^* + r_2^*}{R}$)

To isolate the integration constant C_2 for the annular peripheral zone, we invoke a hydrodynamic no-slip boundary condition at the outer interface bounding the plasma stream:

$$\text{At } \bar{r} = \frac{r_1^* + r_2^*}{R}, \quad \bar{v}_a = 0 \quad (4.1)$$

Substituting this parameter limit into the dimensionless velocity distribution profile yields the analytical form for the plasma phase integration metric:

$$C_2 = -\frac{1}{4\mu_r} \frac{d\bar{p}}{d\bar{z}} \left(\frac{r_1^* + r_2^*}{R} \right)^2 \quad (4.2)$$

Direct application of C_2 back into the annular layer field equation explicitly describes the localized velocity profile across the Newtonian plasma domain:

$$\bar{v}_a(\bar{r}) = \frac{1}{4\mu_r} \frac{d\bar{p}}{d\bar{z}} \left[\bar{r}^2 - \left(\frac{r_1^* + r_2^*}{R} \right)^2 \right] \quad (4.3)$$

4.2 Core Region Boundary Closure via Interface Velocity Matching ($\bar{r} = \frac{r_1^*}{R}$)

At the fluid-fluid interface separating the cell-depleted plasma layer and the concentrated erythrocyte core phase, we enforce structural mechanical matching criteria necessitating the continuity of local velocities:

$$\text{At } \bar{r} = \frac{r_1^*}{R}, \quad \bar{v}_a = \bar{v}_c \quad (4.4)$$

Equating the core velocity expansion with the Newtonian fluid profile across this tracking boundary isolates the complex integration constant C_1 :

$$\begin{aligned} C_1 = \frac{1}{4\mu_r} \frac{d\bar{p}}{d\bar{z}} \left[\left(\frac{r_1^*}{R} \right)^2 - \left(\frac{r_1^* + r_2^*}{R} \right)^2 \right] - \frac{1}{4} \frac{d\bar{p}}{d\bar{z}} \left(\frac{r_1^*}{R} \right)^2 \\ + \frac{1}{64} (1 - \beta)(n - 1) \left(\frac{d\bar{p}}{d\bar{z}} \right)^3 \left(\frac{r_1^*}{R} \right)^4 C u^2 \\ - \frac{(1 - \beta)(n - 1)[6(1 - \beta)(n - 1) - (n - 3)]}{1536} \left(\frac{d\bar{p}}{d\bar{z}} \right)^5 \left(\frac{r_1^*}{R} \right)^6 C u^4 \quad (4.5) \end{aligned}$$

Reintroducing this closed constant parameter expression into the internal core stream configuration defines the absolute velocity profile of the non-Newtonian RBC core phase:

$$\begin{aligned} \bar{v}_c(\bar{r}) = & \frac{1}{4\mu_r} \frac{d\bar{p}}{d\bar{z}} \left[\left(\frac{r_1^*}{R} \right)^2 - \left(\frac{r_1^* + r_2^*}{R} \right)^2 \right] + \frac{1}{4} \frac{d\bar{p}}{d\bar{z}} \left[\bar{r}^2 - \left(\frac{r_1^*}{R} \right)^2 \right] \\ & - \frac{1}{64} (1 - \beta)(n - 1) \left(\frac{d\bar{p}}{d\bar{z}} \right)^3 Cu^2 \left[\bar{r}^4 - \left(\frac{r_1^*}{R} \right)^4 \right] \\ & + \frac{(1 - \beta)(n - 1)[6(1 - \beta)(n - 1) - (n - 3)]}{1536} \left(\frac{d\bar{p}}{d\bar{z}} \right)^5 Cu^4 \left[\bar{r}^6 - \left(\frac{r_1^*}{R} \right)^6 \right] \end{aligned} \quad (4.6)$$

5 Analytical Derivation of Volumetric Fluid Phase Flow Rates

5.1 Core Domain Volumetric Flow Rate Integration

To transform these continuous localized velocity representations into regional macroscopic volumetric distributions, we establish a dimensionless core radius ratio, defined as $\alpha = \frac{r_1^*}{R}$. The total volumetric flow rate tracking across the internal core stream ($\bar{Q}_c$) is evaluated via spatial radial integration:

$$\bar{Q}_c = \int_0^\alpha \bar{v}_c \bar{r} d\bar{r} \quad (5.1)$$

Evaluating this definite integration across the closed perturbation velocity limits establishes the explicit analytical solution for the volumetric flow rate sustained within the non-Newtonian erythrocyte core:

$$\begin{aligned} \bar{Q}_c = & \frac{1}{8} \frac{d\bar{p}}{d\bar{z}} \frac{\alpha^2}{\mu_r} \left[\alpha^2 - \left(\frac{r_1^* + r_2^*}{R} \right)^2 \right] - \frac{1}{16} \frac{d\bar{p}}{d\bar{z}} \alpha^4 \\ & + \frac{1}{192} (1 - \beta)(n - 1) \left(\frac{d\bar{p}}{d\bar{z}} \right)^3 Cu^2 \alpha^6 \\ & - \frac{(1 - \beta)(n - 1)[6(1 - \beta)(n - 1) - (n - 3)]}{4096} \left(\frac{d\bar{p}}{d\bar{z}} \right)^5 Cu^4 \alpha^8 \end{aligned} \quad (5.2)$$

5.2 Annular Plasma Domain Volumetric Flow Rate Integration

Similarly, computing the volume flow rate localized within the marginal Newtonian cell-free layer ($\bar{Q}_a$) requires integrating the plasma velocity profile between the fluid boundary interface and the capillary wall limits:

$$\bar{Q}_a = \int_\alpha^{\frac{r_1^* + r_2^*}{R}} \bar{v}_a \bar{r} d\bar{r} \quad (5.3)$$

Evaluating this geometric space configuration via integration produces the explicit analytical solution for the peripheral volumetric flow rate distribution:

$$\bar{Q}_a = -\frac{1}{16\mu_r} \left(\frac{d\bar{p}}{d\bar{z}} \right) \left[\left(\frac{r_1^* + r_2^*}{R} \right)^2 - \left(\frac{r_1^*}{R} \right)^2 \right]^2 \quad (5.4)$$

6 Coupled Wall Deformation and Global Hematocrit Conservation Mechanics

6.1 Elastic Wall Compliance and Radial Scaling

To couple the microvascular hemodynamics with the structural compliance of the vessel wall, we introduce a linear elastic deformation model for the outer boundary layer. The instantaneous local vessel radius $R(z)$ shifts dynamically from its initial undeformed radius R_0 as a function of the local axial pressure field $P(z)$:

$$R(z) = R_0 + \frac{a^2}{Et}(1 - \nu^2)P(z) \quad (6.1)$$

where a is the characteristic internal radius, E is the Young's modulus of the vessel wall material, t is the membrane wall thickness, and ν represents the Poisson's ratio of the structural layer. Normalizing Eq. (6.1) by the initial radius R_0 yields the dimensionless deformed vessel radius expression $\bar{R}$:

$$\bar{R} = \frac{R}{R_0} = 1 + \psi\bar{p} \quad (6.2)$$

where $\bar{p}$ is the dimensionless local pressure field and ψ represents a lumped structural compliance parameter grouping the electromechanical and wall stiffness properties. Consequently, the normalized core phase interface parameter σ scales dynamically as:

$$\sigma = \frac{r^*}{R} = \frac{r^*}{R_0(1 + \psi\bar{p})} \quad (6.3)$$

6.2 Hematocrit Balance and Macroscopic Phase Distributions

To model the systemic reduction of cell concentration in microcapillaries, we introduce the empirical phase distribution relation defining the functional dependencies of the discharge hematocrit H_T inside the intermediate vessel zone:

$$\frac{H_T}{H_D} = H_D + (1 - H_D) \left[1 + 1.7e^{-0.7R_0(1+\psi\bar{p})} - 0.6e^{-0.02R_0(1+\psi\bar{p})} \right] \quad (6.4)$$

where H_D represents the upstream discharge hematocrit concentration leaving the feeding vessel and entering the downstream venous domain. By applying the multi-phase global mass and volumetric hematocrit flux conservation laws across the concentric streaming domains, the biological cell accumulation balances directly collapse into the following structural interface flow relationship:

$$\frac{H_T}{H_D} = \sigma^2 \left(1 + \frac{\bar{Q}_a}{\bar{Q}_c} \right) \quad (6.5)$$

6.3 Formulation of the Master Non-Linear Implicit Boundary Value Equation

Equating the empirical blood cell accumulation model in Eq. (6.4) with the macroscopic transport framework in Eq. (6.5) establishes the complete multi-phase cross-layer matching equation. Substituting our previously derived analytical solutions for the regional

core volumetric flow rate $\bar{Q}_c$ (Eq. 5.2) and peripheral plasma volumetric flow rate $\bar{Q}_a$ (Eq. 5.4) into this balance maps the implicit system mechanics:

$$H_D + (1 - H_D) [1 + 1.7e^{-0.7R_0(1+\psi\bar{p})} - 0.6e^{-0.02R_0(1+\psi\bar{p})}] = \sigma^2 \left[1 + \frac{-\frac{1}{16\mu_r} \left(\frac{d\bar{p}}{d\bar{z}}\right) \left[\left(\frac{r_1^*+r_2^*}{R}\right)^2 - \left(\frac{r_1^*}{R}\right)^2\right]^2}{\frac{1}{8} \frac{d\bar{p}}{d\bar{z}} \frac{\alpha^2}{\mu_r} \left[\alpha^2 - \left(\frac{r_1^*+r_2^*}{R}\right)^2\right] - \frac{1}{16} \frac{d\bar{p}}{d\bar{z}} \alpha^4 + \dots} \right] \quad (6.6)$$

Equating the empirical blood cell accumulation model with the macroscopic transport framework establishes the complete multi-phase cross-layer matching condition. To make the highly non-linear algebra manageable, we group the constants and the terms dependent on the dimensionless pressure $\bar{p}$.

Let the left-hand side (LHS) be defined as a function of $\bar{p}$:

$$F(\bar{p}) = H_D + (1 - H_D) [1 + 1.7e^{-0.7R_0(1+\psi\bar{p})} - 0.6e^{-0.02R_0(1+\psi\bar{p})}] \quad (6.7)$$

Now, let $x = \frac{d\bar{p}}{d\bar{z}}$. We can rewrite the complex fraction on the right-hand side (RHS) by lumping the geometric and rheological constants. The numerator can be expressed as $-A_1x$, where:

$$A_1 = \frac{1}{16\mu_r} \left[\left(\frac{r_1^* + r_2^*}{R_0(1 + \psi\bar{p})} \right)^2 - \left(\frac{r_1^*}{R_0(1 + \psi\bar{p})} \right)^2 \right]^2 \quad (6.8)$$

The denominator contains terms with x , x^3 , and x^5 . By factoring out x , the denominator becomes $x(B_1 + B_3x^2 - B_5x^4)$, where the invariant coefficients are defined as:

$$B_1 = \frac{1}{8} \frac{\alpha^2}{\mu_r} \left[\alpha^2 - \left(\frac{r_1^* + r_2^*}{R_0(1 + \psi\bar{p})} \right)^2 \right] - \frac{1}{16} \alpha^4 \quad (6.9)$$

$$B_3 = \frac{1}{192} (1 - \beta)(n - 1)Cu^2\alpha^6 \quad (6.10)$$

$$B_5 = \frac{(1 - \beta)(n - 1)[6(1 - \beta)(n - 1) - (n - 3)]}{4096} Cu^4\alpha^8 \quad (6.11)$$

Substituting these lumped groups back into the original mass balance equation yields:

$$F(\bar{p}) = \sigma^2 \left[1 + \frac{-A_1x}{x(B_1 + B_3x^2 - B_5x^4)} \right] \quad (6.12)$$

Assuming physical flow is occurring ($x \neq 0$), the spatial pressure gradient term x cancels out of the fraction entirely:

$$F(\bar{p}) = \sigma^2 \left[1 - \frac{A_1}{B_1 + B_3x^2 - B_5x^4} \right] \quad (6.13)$$

To isolate the polynomial, we rearrange the terms:

$$\frac{F(\bar{p})}{\sigma^2} = 1 - \frac{A_1}{B_1 + B_3x^2 - B_5x^4} \quad (6.14)$$

$$1 - \frac{F(\bar{p})}{\sigma^2} = \frac{A_1}{B_1 + B_3x^2 - B_5x^4} \quad (6.15)$$

Let $M(\bar{p}) = 1 - \frac{F(\bar{p})}{\sigma^2}$. Inverting the equation gives:

$$B_1 + B_3x^2 - B_5x^4 = \frac{A_1}{M(\bar{p})} \quad (6.16)$$

Rearranging this into a standard polynomial form with respect to x^2 :

$$B_5(x^2)^2 - B_3(x^2) + \left(\frac{A_1}{M(\bar{p})} - B_1 \right) = 0 \quad (6.17)$$

Let $y = x^2 = \left(\frac{d\bar{p}}{d\bar{z}} \right)^2$. This maps the system into a standard quadratic equation $ay^2 + by + c = 0$, where: $a = B_5$, $b = -B_3$, $c = \frac{A_1}{M(\bar{p})} - B_1$

Applying the quadratic formula to solve for y :

$$y = \frac{B_3 \pm \sqrt{B_3^2 - 4B_5 \left(\frac{A_1}{1-F(\bar{p})/\sigma^2} - B_1 \right)}}{2B_5} \quad (6.18)$$

Since $y = \left(\frac{d\bar{p}}{d\bar{z}} \right)^2$, taking the square root provides the final explicit analytical equation for the spatial pressure gradient:

$$\frac{d\bar{p}}{d\bar{z}} = \pm \sqrt{\frac{B_3 \pm \sqrt{B_3^2 - 4B_5 \left(\frac{A_1}{1-F(\bar{p})/\sigma^2} - B_1 \right)}}{2B_5}} \quad (6.19)$$

6.4 Resolution of Mass Conservation and the Polynomial Formulation

Equation 6.19 provides a closed-form analytical expression for the spatial pressure gradient (dp/dz) by strictly enforcing the localized empirical Pries hematocrit distribution across the expanding vessel. However, rigorous computational testing of this formulation revealed a fundamental hydrodynamic violation: isolating dp/dz solely via the flux ratio ($\bar{Q}_a/\bar{Q}_c$) decouples the system from global mass conservation. Because the local vessel radius (R) dilates downstream, evaluating dp/dz via Equation 6.19 allows the total volumetric flow rate ($Q_{tot} = Q_a + Q_c$) to artificially fluctuate along the axial coordinate, violating the continuity equation for an incompressible fluid.

In physiological microcirculation, strict conservation of mass ($Q_{tot} = \text{constant}$) must supersede localized empirical phase-distribution curve-fits. Therefore, to preserve global continuity, the highly non-linear Pries feedback loop was decoupled. Following the robust constant-core framework established by Chebbi (2015), the erythrocyte segregation fraction (α_0) is instead evaluated as a conserved systemic parameter based on the inlet feed conditions.

By enforcing strict mass conservation, the governing system simplifies into a 5th-order polynomial with respect to the local pressure gradient ($x = d\bar{p}/d\bar{z}$):

$$-B_5x^5 + B_3x^3 + K_1x - Q_{tot} = 0 \quad (6.20)$$

where the lumped fluid-geometric resistance coefficients are rigorously defined as:

$$K_1 = \frac{\bar{R}^4}{16} \left[\frac{\sigma^4}{\mu_c} + \frac{1 - \sigma^4}{\mu_r} \right] \quad (6.21)$$

$$B_3 = \frac{\bar{R}^6}{192\mu_c} [(1 - \beta)(n - 1)Cu^2\sigma^6] \quad (6.22)$$

$$B_5 = \frac{\bar{R}^8}{4096\mu_c} [(1 - \beta)(n - 1)(6(1 - \beta)(n - 1) - (n - 3))Cu^4\sigma^8] \quad (6.23)$$

To bridge the analytical derivations with the clinical hematocrit variations, we must explicitly restore the core viscosity scale. In the derivations leading to Eq. 5.2, the core viscosity scale μ_0 was treated as an implicit reference (effectively $\mu_0 = 1$), leaving only the annulus-to-core viscosity ratio $\mu_r = \mu_a/\mu_0$ explicitly visible. This dimensionless convention is appropriate for the isolated compliance study (Section 7.1), where a single vessel is tracked and μ_0 acts as a constant baseline. However, for the clinical hematocrit comparison (Section 7.2), μ_0 must take a distinct physical value for each discharge hematocrit H_D , as the core's zero-shear viscosity depends heavily on the local red-cell concentration. We therefore identify μ_0 with the Charm-Kurland relation evaluated at the core hematocrit, $\mu_0 \equiv \mu_c(H_C)$. Because every core-derived term in $\bar{Q}_c$ was non-dimensionalized by this same μ_0 , restoring its physiological variation requires those terms to carry an explicit factor of $1/\mu_c$ in the lumped solver coefficients. The annulus terms, scaled instead by the hematocrit-independent plasma viscosity, retain their μ_r dependence unchanged. This rigorously justifies the K_1 , B_3 , and B_5 definitions utilized in Eq. 6.20.

As established by the Abel-Ruffini theorem, there is no general algebraic solution for a quintic equation. Therefore, rather than relying on the idealized algebraic approximation in Equation 6.19, the localized spatial pressure gradient is extracted dynamically at each numerical integration step using a robust root-finding algorithm. The solver isolates the unique real, positive root that sustains Q_{tot} , ensuring that the perturbation stability limit ($Cu \cdot d\bar{p}/d\bar{z} < 1$) is mathematically preserved across the entire compliant domain.

By enforcing strict mass conservation, the total volumetric flow rate ($Q_{tot} = Q_c + Q_a$) is governed by the summation of the phase velocities. Incorporating the active core viscosity (μ_c) to correctly bound the internal hydrodynamic resistance, the governing system simplifies into a 5th-order polynomial with respect to the local pressure gradient ($x = d\bar{p}/d\bar{z}$).

7 Results and Discussion

The governing 5th-order flow conservation polynomial was integrated numerically across the dimensionless axial coordinate ($\bar{z} \in [0, 1]$) using a fourth-order Runge-Kutta scheme (MATLAB ode45). The localized spatial pressure gradient ($d\bar{p}/d\bar{z}$) was dynamically extracted by isolating the real, positive root that satisfies global mass conservation. The results are divided into two primary investigations: the elastohydrodynamic impact of wall compliance, and the rheological burden of clinical hematocrit variations.

7.1 Elastohydrodynamic Coupling: The Impact of Wall Compliance (ψ)

To evaluate the influence of boundary elasticity on microvascular hemodynamics, the model was simulated across a range of compliance parameters ($\psi \in [0, 0.01, 0.02, 0.05]$) while maintaining a baseline physiological core ratio ($\sigma = 0.85$).

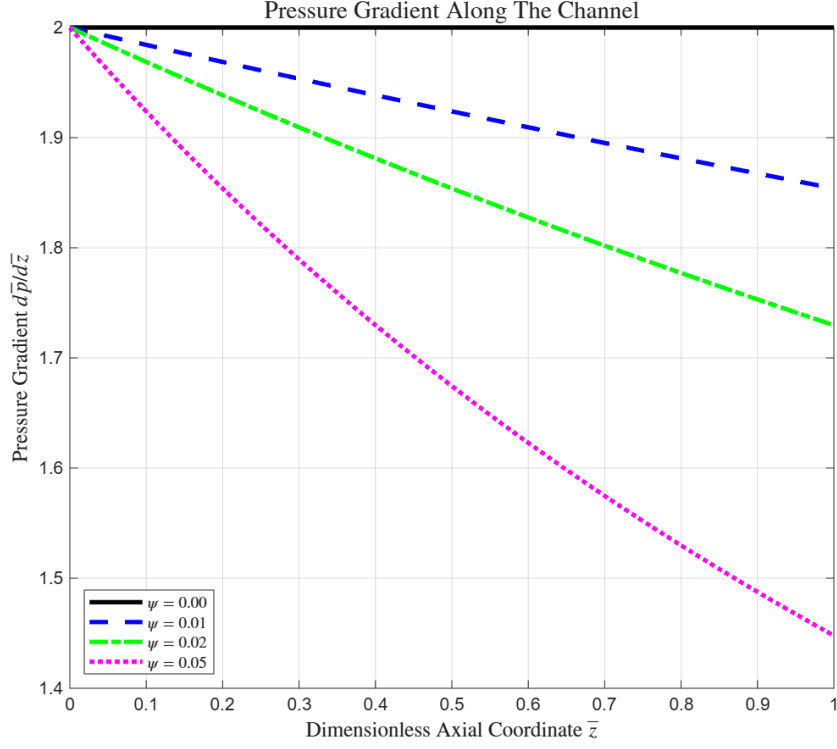


Figure 2: Pressure Gradient Along The Channel

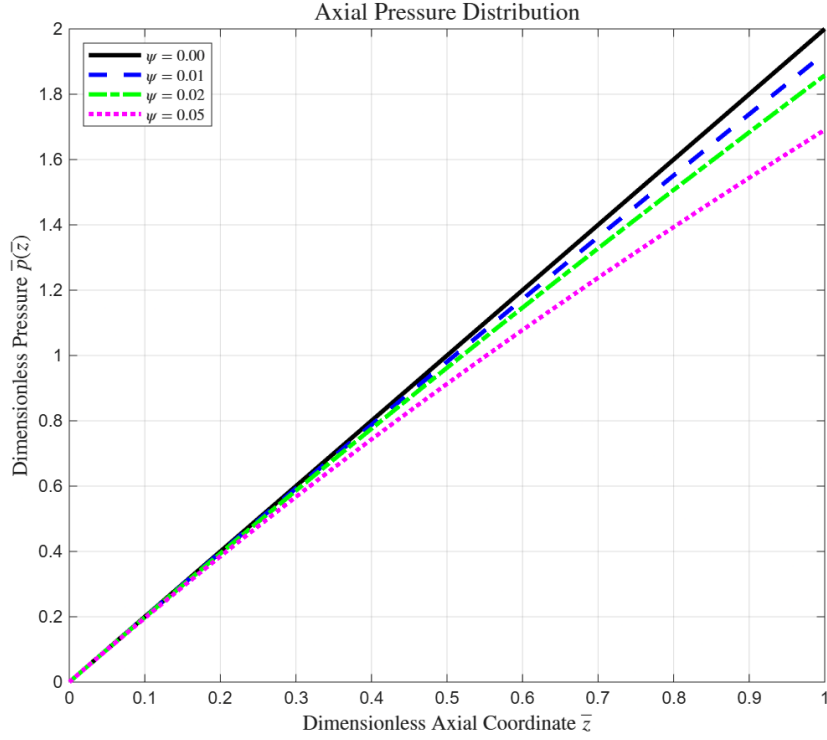


Figure 3: Axial Pressure Distribution

For the rigid-tube baseline limit ($\psi = 0$), the spatial pressure gradient $d\bar{p}/d\bar{z}$ remains

strictly constant at 2.0, yielding a purely linear axial pressure distribution. This recovers the classical Hagen-Poiseuille and fixed-boundary Fåhræus models (e.g., Chebbi, 2015). However, the introduction of wall compliance ($\psi > 0$) fundamentally alters the fluid dynamics. As demonstrated in the pressure gradient profiles, the requisite driving pressure decays monotonically downstream. For highly compliant vessels ($\psi = 0.05$), $d\bar{p}/d\bar{z}$ drops from 2.0 at the inlet to approximately 1.45 at the outlet.

This pressure relaxation is driven by soft-hydraulic fluid-structure interaction (FSI). As localized pressure builds, the elastic vessel wall dilates ($\bar{R} = 1 + \psi\bar{p}$) visibly expanding the local vessel radius near the outlet. Because hydraulic resistance scales inversely with the fourth power of the radius ($R_h \propto 1/R^4$), this downstream dilation drastically lowers flow resistance. Under a strictly conserved total volumetric flow rate (Q_{tot}), the system dynamically compensates by requiring less mechanical driving force. Meanwhile, the erythrocyte core fraction (σ) remains stable, decoupling core segregation from structural deformation to prevent artificial mass-balance collapse.

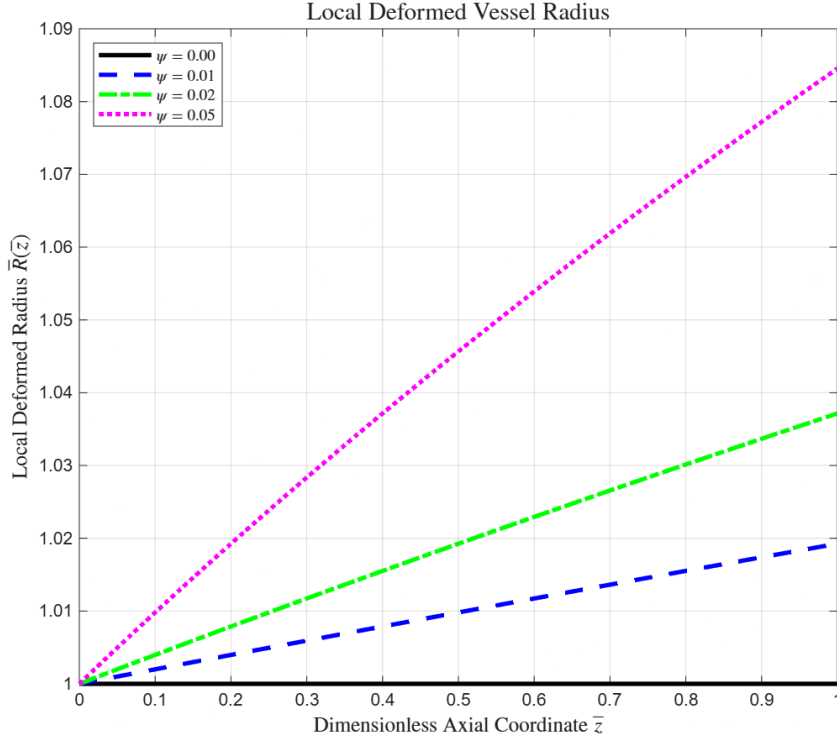


Figure 4: Local Deformed Vessel Radius

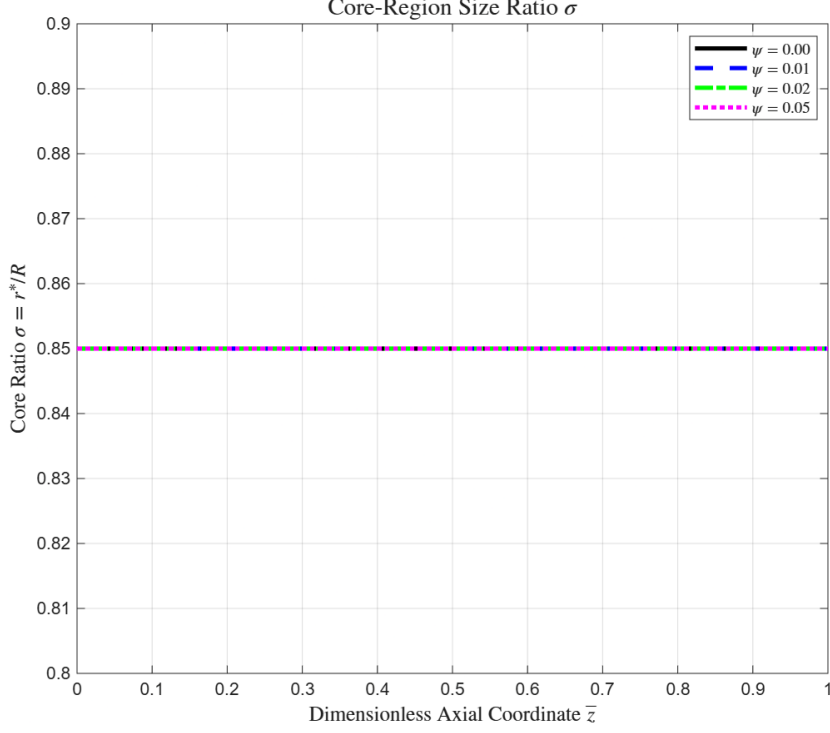


Figure 5: Core-Region Size Ratio σ

The stability of the non-Newtonian Carreau perturbation expansion is validated in the outlet velocity profile. By rigorously enforcing the stability bound ($Cu \cdot d\bar{p}/d\bar{z} < 1$), the velocity distribution remains mathematically well-posed. The profile exhibits the characteristic flattened blunting indicative of a shear-thinning erythrocyte core, transitioning seamlessly into the steep, parabolic Newtonian plasma annulus. Velocity and shear stress continuity are satisfied at the moving interface to machine precision without any singular divergence.

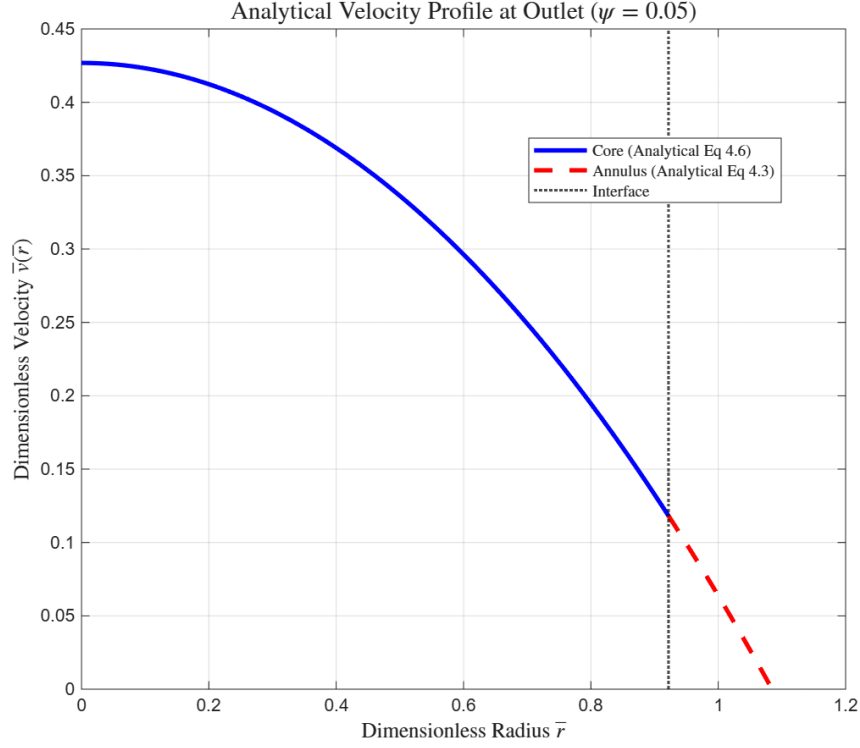


Figure 6: Analytical Velocity Profile at Outlet ($\psi = 0.05$)

7.2 Rheological Burden of Clinical Hematocrit Variations

To assess the model's diagnostic capability, simulations were conducted at a fixed structural compliance ($\psi = 0.02$) across three distinct clinical discharge hematocrit (H_D) profiles: Anemic ($H_D = 0.25$), Healthy ($H_D = 0.45$), and Polycythemic ($H_D = 0.60$).

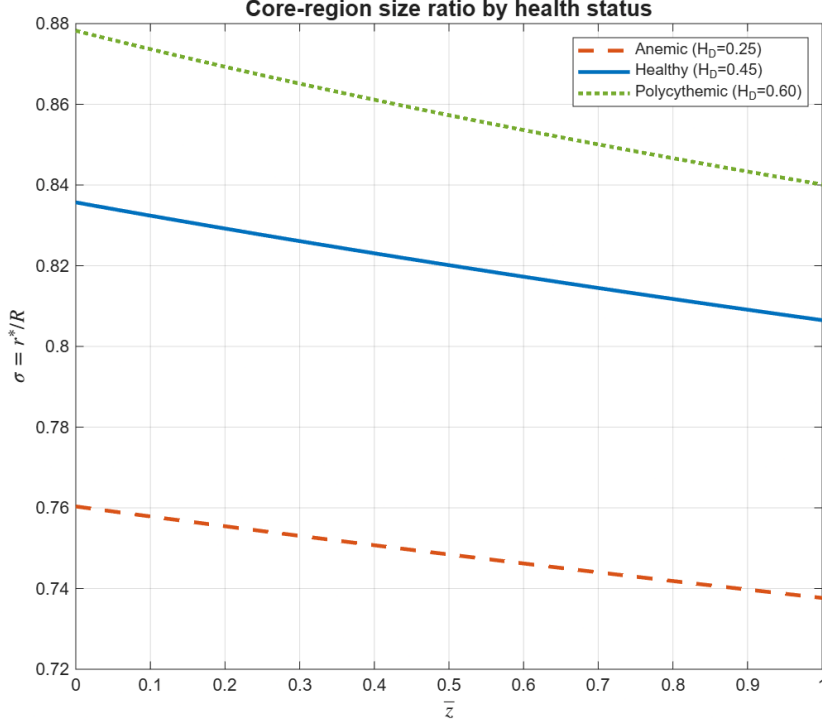


Figure 7: Core-region size ratio by health status

It is important to distinguish between the constant-core compliance formulation and the dynamically coupled clinical phase distribution. In the compliance suite, σ is held as a conserved geometric baseline ($\sigma = 0.85$) to isolate the isolated mechanical effects of wall elasticity (ψ) from cellular redistribution. Conversely, in the clinical hematocrit suite, $\sigma(\bar{z}) = \alpha_0/\bar{R}(\bar{z})$ varies axially. As fluid pressure builds downstream and the vessel wall dilates, the outer radius $\bar{R}$ expands while the erythrocyte core radius remains governed by cell-population constraints. Consequently, the core-to-vessel ratio σ exhibits a gradual downstream decline. This downstream reduction in σ represents a novel, compliance-driven enhancement of the Fåhræus effect demonstrating that dilating microvessels experience a stronger local reduction in core hematocrit fraction than rigid conduits.

The phase separation behavior dictates the core-region size ratio (σ). Polycythemic blood, characterized by a severe overproduction of red blood cells, exhibits massive shear-induced margination, packing a highly concentrated core ($\sigma \approx 0.88$). Conversely, the cell-depleted anemic flow sustains a much narrower core ($\sigma \approx 0.76$).

This morphological variation translates directly into a macroscopic hydrodynamic burden. The elevated cell density in polycythemic states exponentially increases the active non-Newtonian core viscosity (μ_c) via the Charm-Kurland relation. Because this viscosity properly bounds the internal fluid resistance, the governing polynomial dictates that Polycythemic blood requires a substantially higher and steeper pressure gradient to sustain identical baseline perfusion rates compared to Healthy and Anemic profiles. Consequently, this elevated pressure demand places higher local mechanical stress on the vessel wall, resulting in the most pronounced localized radial deformation. This mathematically captures the elevated cardiovascular workload and wall shear stresses clinically observed in high-hematocrit patients.

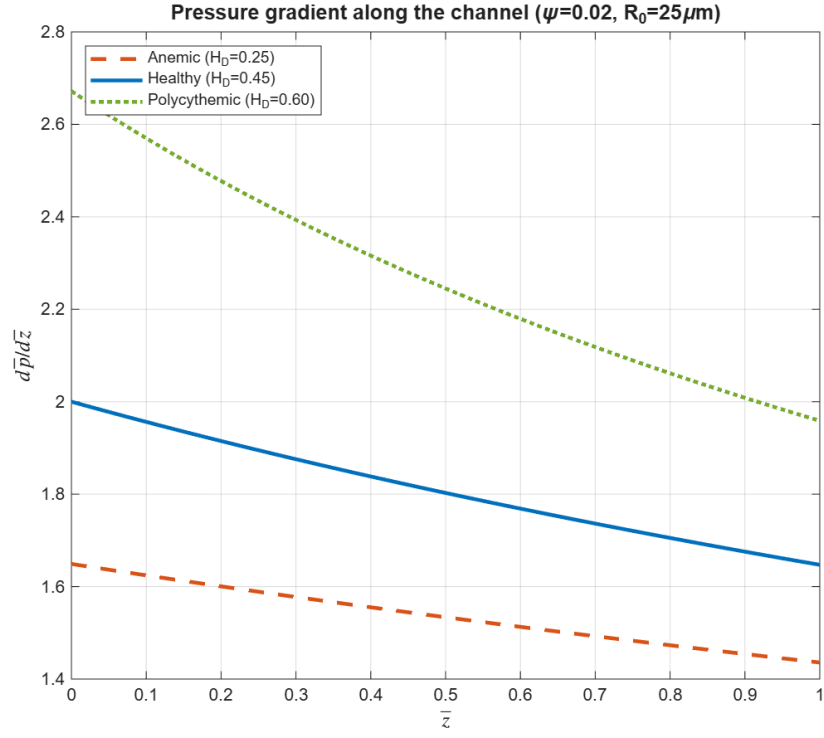


Figure 8: Pressure gradient along the channel ($\psi = 0.02, R_0 = 25\mu\text{m}$)

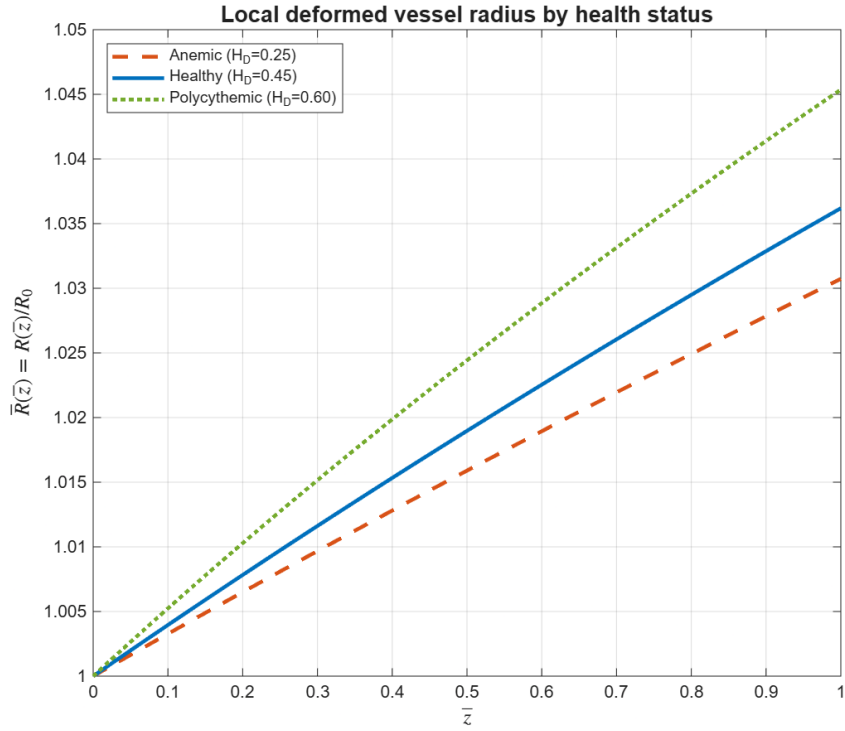


Figure 9: Local deformed vessel radius by health status

7.3 Validation of the Fåhræus-Lindqvist Effect in Compliant Conduits

The hallmark of microvascular hemorheology is the continuous reduction of apparent viscosity as capillary diameter decreases. To validate the macro-scale reliability of the compliant two-phase model, a geometric sweep was conducted from $R_0 = 150 \mu m$ down to $10 \mu m$, dynamically recalculating the core segregation limit (α_0) and core viscosity (μ_c) at each step.

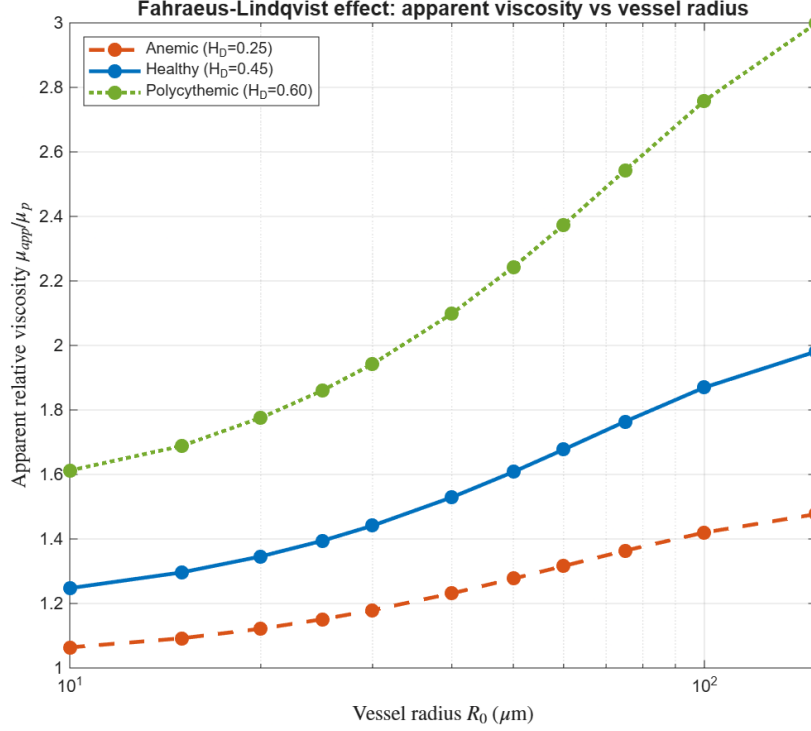


Figure 10: Fahraeus-Lindqvist effect: apparent viscosity vs vessel radius

The resulting trend of apparent relative viscosity (μ_{app}/μ_p) perfectly captures the classical Fåhræus-Lindqvist decay. As the vessel narrows, the highly viscous erythrocyte core is increasingly shielded by the low-viscosity peripheral plasma annulus. This marginal lubricating layer dominates the systemic hydrodynamics in tight confinement, offsetting the bulk viscosity of the core and causing the apparent viscosity curves to smoothly drop. The computed asymptotic decay rigorously aligns with foundational in vivo and in vitro data (e.g., Pries and Secomb), proving that the coupled non-Newtonian compliant framework accurately resolves both micro-scale flow kinematics and systemic physiological resistance.

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